

QUOTA — Phases 0–8

Window: **1 Oct** → **13 Oct** (submit a day early; deadline 14 Oct 09:29 IST). Phases overlap, and dates are targets. Each phase ends with an **exit check** you can run, not a feeling.

Cut order if time runs out: Cleanverse → Nansen → Dynamic. **Never cut:** passkey custody, RLN proofs, slash safety path, Privy, Qwen agent, docs, demo video.

Phase 0 — Gates and setup (1–2 Oct)

Goal: kill unknowns before writing real code.

- ☐ Monorepo, `.env.example`, public GitHub repo, grant `metropolis@hackathon.monad.xyz`.
- ☐ Foundry + Node toolchain; deploy a hello contract to Monad testnet.
- ☐ Confirm testnet chain ID, RPC, faucet, explorer.
- ☐ **Gate BTX:** ask Monad mentors; find the interface; does it hide target + calldata or only payload? Decide `btx` vs commit–reveal as primary.
- ☐ **Gate Privy:** `personal_sign` deterministic? policy covers what? sign-only works on Monad?
- ☐ **Gate Dynamic:** one signed Monad tx, sign-only; delegated-access webhook reachable. Fail → drop.
- ☐ **Gate Cleanverse:** invitation code, read the 3 guides, CVA on testnet, CVI callable on-chain. Fail → drop.
- ☐ Nansen key and one Profiler call returning data for a Monad address.
- ☐ Qwen key, model ID, one tool-calling round trip.
- ☐ Start integrator outreach (parallel, continues to Phase 8).
- ☐ Community group decision.

Exit: a `docs/gates.md` listing each gate PASS/FAIL with evidence. Scope is now fixed from it.

Phase 1 — Contracts: custody and registry (2–4 Oct)

- ☐ `PasskeyAuth`: WebAuthn assertion verification via `0x100` (all checks in PRD C2).
- ☐ Test vectors: valid, wrong challenge, wrong origin, wrong rpld, missing UP/UV, high-s, replayed nonce, malformed JSON.
- ☐ `QuotaRegistry`: tree (Poseidon), `registerPasskey`, `enroll`, `topUp`, `requestUnstake` / `unstake`, recent-roots window.
- ☐ Measure enroll gas on Monad; apply risk fallback if too costly.
- ☐ Deploy to testnet; verify on explorer.

Exit: `forge test` green incl. all negative cases; a real browser passkey assertion verifies on-chain on testnet.

Phase 2 — ZK: RLN proofs (3–6 Oct)

- ☐ Vendor and checksum RLN-v2 artifacts; pick tree depth.
- ☐ `@quota/client` prover: `signRequest` with epoch/ `k` tracking.
- ☐ `@quota/server` verifier + nullifier store + secret recovery from two shares.
- ☐ Recent-roots sync from the registry.
- ☐ Benchmark proof gen/verify and record real numbers.

Exit: script: 3 honest requests verify; a 2nd use of the same `k` recovers `a0` exactly; per-server external nullifier prevents cross-server collisions.

Phase 3 — Slash and BTX path (5–7 Oct)

- ☐ `slash(a0, limit, receiver)` with leaf recompute and payout.
- ☐ `SubmitPath`: `commitReveal` (guaranteed) and `btx` (if gate passed).
- ☐ **Searcher test:** a bot watches the public path and tries to copy the slash. It must fail on both paths where the path is private/committed, and succeed on a naive path (to prove the race is real).
- ☐ Slasher service: violation → queue → slash.

Exit: on-chain slash with reward received; copy-and-steal test result documented.

Phase 4 — SDKs, middleware, demo MCP server (6–8 Oct)

- ☐ `@quota/server` middleware for Express/Hono and an MCP wrapper.
- ☐ `@quota/slasher` with the wallet adapter interface; `LocalKeyWallet` first.
- ☐ Reference MCP server (`apps/demo-mcp`) protected by QUOTA.
- ☐ Quickstarts for both personas; measure a fresh developer's integration time.

Exit: a stranger (or a clean machine) follows the service quickstart in ≤ 10 minutes.

Phase 5 — Wallet integrations (7–9 Oct)

- ☐ **Privy:** operator login, server wallet with owner = operator user, runtime key as additional signer with override policies, secret derivation from signature, sign-only.
- ☐ **Dynamic:** service login, embedded wallet for rewards, delegated access to the slasher, or server-wallet fallback. (Skip if Phase 0 failed.)
- ☐ Both through `wallets` adapters. Contract and SDK code must not depend on either provider.
- ☐ Negative test: policy blocks a transaction to any contract other than the registry.

Exit: the end-to-end flow runs with Privy on the agent side and Dynamic on the service side, no private keys in `.env` except sponsor auth keys.

Phase 6 — Trust tiers (8–10 Oct) — conditional

- ☐ **Nansen Screener:** profile a new operator wallet (labels, related wallets, funding, counterparties), sign attestation, Screened tree accepts it; recidivism rule against slashed wallets.
- ☐ Service console shows tree choice, cluster/concentration stats, slash history.
- ☐ **Cleanverse Compliant tree** (only if the gate passed): CVA as stake, CVI check before every CVA movement, including slash payout.
- ☐ "Remove it and it breaks" check for each: write one test where the tier fails closed without the sponsor component.

Exit: a screened enrollment succeeds for a clean wallet and is refused for a related-to-slashed wallet; (if CVI) a non-verified wallet cannot receive CVA. If neither works by 10 Oct, cut and move on.

Phase 7 — Scout agent, consoles, deploy (9–11 Oct)

- ☐ **Qwen Scout:** tool-calling loop (`quota_status` , `call_tool` , `topup_stake` , `switch_server`), plans a research task across ≥ 2 QUOTA servers inside quota.
- ☐ Violation script for the slash demo.
- ☐ Operator console + service console deployed with public URLs and test credentials for judges.
- ☐ Contracts verified; README access instructions.
- ☐ Draft the Qwen **article** from the build log.

Exit: a judge with only the README can log in, enroll, watch Scout work, and see a slash.

Phase 8 — Hardening, media, submission (11–13 Oct)

- ☐ Security pass: secrets scan of repo and git history, passkey checks reread, replay tests, `forge coverage` on custody and slash paths.
- ☐ Threat model doc + "why not roll your own" page.
- ☐ **Technical demo video (≤ 3 min, live product, no slides):** enroll with passkey → Scout makes anonymous requests → violation → slash lands. Show the tx.
- ☐ **Pitch video (≤ 2 min):** team, problem, why.
- ☐ Optional 30 s ad. Logo/graphic (JPG/PNG/WEBP ≤ 3 MB).
- ☐ Per-bounty descriptions: how each sponsor is used (see `00-bounty-validation.md`); Dynamic/Privy/Nansen/Cleanverse/Qwen videos where optional.
- ☐ Publish the Qwen article; link it.
- ☐ Record the integrator evidence (a PR, a running URL, or a message).
- ☐ Submit by the evening of 13 Oct. Leave a buffer for portal problems.

Exit: every deliverable in the checklist below is ticked and its URL opens in a private window.

Submission checklist

- ☐ Public repo (access for `metropolis@hackathon.monad.xyz`)
- ☐ Live product on Monad + access instructions + test credentials
- ☐ Technical demo video ≤ 3 min · Pitch video ≤ 2 min · Logo ≤ 3 MB
- ☐ Track 4 selected, plus bounties: Privy, Dynamic, Qwen, Nansen, Cleanverse, Community (*= only if kept after gates)
- ☐ Qwen article URL
- ☐ No secrets in repo or history
- ☐ Named integrator evidence

Day plan at a glance

Day	1–2 Oct	2–4	3–6	5–7	6–8	7–9	8–10	9–11	11–13
Phase	0	1	2	3	4	5	6	7	8